

Common Prefix Storage in Tries

1 Main Idea

Prefix Sharing

A trie stores a common prefix once. Every word that begins with that prefix reuses the same root-to-prefix path.

This is the central space-saving idea of tries. If several strings begin with the same characters, those characters are not copied into separate paths.

2 Example: cat and cats

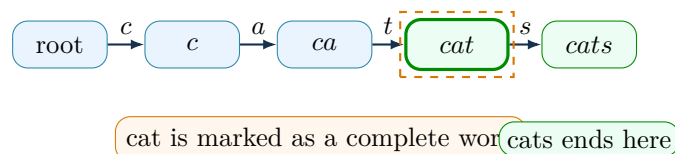
The words

cat and cats

share the prefix

cat.

The longer word continues from the same node where the shorter word ends.



Important Implementation Detail

The node for cat must be marked as a word-ending node even though it has a child for cats. A node can be both a complete word and an internal prefix node.

3 Why Sharing Saves Space

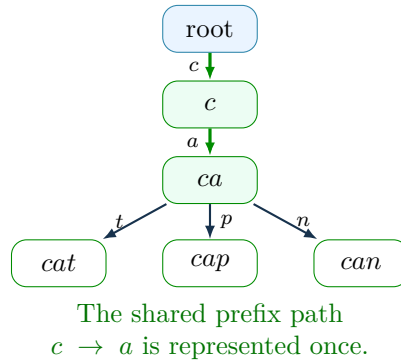
Without a trie, the strings

cat, cap, can

store the prefix ca three times:

Word	Stored prefix	Last character
cat	ca	<i>t</i>
cap	ca	<i>p</i>
can	ca	<i>n</i>

With a trie, the prefix ca is stored once:



4 Character Count Comparison

For

$$S = \{\text{cat}, \text{cap}, \text{can}\},$$

storing the words separately uses

$$|\text{cat}| + |\text{cap}| + |\text{can}| = 3 + 3 + 3 = 9$$

character occurrences.

In the trie, the distinct prefix nodes are

$$c, \quad ca, \quad cat, \quad cap, \quad can.$$

So this example uses 5 non-root prefix/word nodes instead of 9 separate character occurrences.

Savings

The saved work comes from not repeating the shared prefix *ca* for every word.

5 When Prefix Sharing Helps Most

Prefix sharing is strongest when many strings have long common prefixes.

Word set	Effect in a trie
{cat, cap, can}	Strong sharing of prefix <i>ca</i> .
{interview, internet, internal}	Strong sharing of prefix <i>inter</i> .
{cat, dog, sun}	Little sharing because first characters differ.

6 Search Benefit

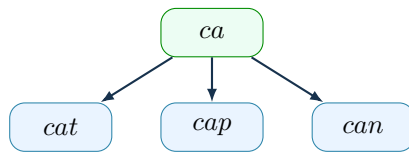
Prefix sharing also helps search. Once the prefix path is found, all words below that node share the prefix.

For example, after reaching

ca,

the subtree below that node contains exactly the words beginning with *ca*:

cat, cap, can.



A prefix query can return the whole subtree below the matched prefix.